INCH-POUND

MIL-S-19500/508A
30 November 1993
SUPERSEDING
MIL-S-19500/508(ER)
15 September 1975

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, POWER
TYPES 2N6437 AND 2N6438, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, power transistors. Two levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Type	P_T 1/		V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{OP} and T_{STG}	$R_{\theta JC}$
	T_C = +25°C	T_C = +100°C							
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N6437	200	112	120	100	6	25	10	-55 TO +200	.875 max
2N6438	200	112	140	120	6	25	10	-55 TO +200	.875 max

1/ Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 1.14 W/°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/	h_{FE2} 1/	$V_{CE(sat)2}$	$V_{CE(sat)1}$	$V_{BE(sat)1}$	Switching	
Limits	V_{CE} = 2 V dc I_C = 25 A dc	V_{CE} = 2 V dc I_C = 10 A dc	I_C = 25 A dc I_B = 2.5 A dc	I_C = 10 A dc I_B = 1.0 A dc	I_C = 10 A dc I_B = 1.0 A dc	t_{on}	t_{off}
			V dc	V dc	V dc	µs	µs
Min	12	30					
Max		120	1.8	1.0	1.8	0.5	1.25

	C_{obo} V_{CB} = 10 V dc I_E = 0 .1 MHz ≤ f ≤ 1 MHz	$\lvert h_{fe} \rvert$ V_{CE} = 10 V dc I_C = 1 A dc f = 10 MHz
	pF	
Min		4
Max	700	12

1/ Pulsed, (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory and Power Sources Directorate, ATTN: AMSRL-EP-RD, Ft. Monmouth, NJ 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 12, T-5A.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements
	JANTX and JANTXV levels
11	I_{CEX1} and h_{FE1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or 2 µA dc whichever is greater. Δh_{FE1} = ±25% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ± 12.5°C; V_{CB} ≥ 20 V dc.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

Subgroup	Method	Condition
B3	1037	V_{CB} ≥ 20 V dc; ΔT_J between cycles ≥ 100°C. T_{on} = T_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	3131	See 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

Subgroup	Method	Condition
C6	1037	V_{CB} ≥ 20 V dc; ΔT_J between cycles ≥ 100°C. T_{on} = T_{off} = 3 minutes minimum for 6,000 cycles. No heat sink or forced-air cooling on device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 50 mA dc, Pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6437				100		V dc
2N6438				120		V dc
Collector to emitter cutoff current	3041	Bias condition D,	I_{CEO}		50	µA dc
2N6437		V_{CE} = 50 V dc				
2N6438		V_{CE} = 60 V dc				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		100	µA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc,	I_{CEX1}		5.0	µA dc
2N6437		V_{CE} = 100 V dc				
2N6438		V_{CE} = 120 V dc				
Collector to base cutoff current	3036	Bias condition D,	I_{CBO}		10	µA dc
2N6437		V_{CB} = 120 V dc				
2N6438		V_{CB} = 140 V dc				
Base emitter voltage (saturated)	3066	Test condition A, I_C = 10 A dc, I_B = 1.0 A dc, Pulsed (see 4.5.1)	$V_{BE(sat)}$		1.8	V dc
Collector to emitter saturated voltage	3071	I_C = 10 A dc, I_B = 1.0 A dc, Pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
Collector to emitter saturated voltage	3071	I_C = 25 A dc, I_B = 2.5 A dc, Pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.8	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = .5 A dc, Pulsed (see 4.5.1)	h_{FE1}	40		

See footnote at end of table.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 10 A dc, Pulsed (see 4.5.1)	h_{FE2}	30	120	
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 25 A dc Pulsed (see 4.5.1)	h_{FE3}	12		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX2}		1.0	mA dc
2N6437		V_{CE} = 100 V dc				
2N6438		V_{CE} = 120 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 10 A dc, Pulsed (see 4.5.1)	h_{FE4}	10		
Subgroup 4						
Pulse response		Test condition A except test circuit and pulse requirements per fig. 1 herein				
Turn-on time		V_{CC} = 80 V dc, I_C = 10 A dc, I_{B1} = 1.0 A dc	t_{on}		0.5	µsec
Turn-off time		V_{CC} = 80 V dc, I_C = 10 A dc, I_{B1} = I_{B2} = 1.0 A dc	t_{off}		1.25	µsec
Storage time		V_{CC} = 80 V dc, I_C = 10 A dc, I_{B1} = I_{B2} = 1.0 A dc	t_s		1.0	µsec
Magnitude of common-emitter small-signal current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 1 A dc,	$\|h_{fe}\|$	4.0	12	
Output capacitance (open circuit)	3236	V_{CB} = 10 V dc, I_E = 0, 0.1 MHz ≤ f ≤ 1 MHz	C_{obo}		700	pF

See footnote at end of table.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area continuous (dc operation)	3051	T_C = 25°C, t = 1s, 1 cycle, (see figure 2)				
Test 1 (Both device types)		V_{CE} = 8 V dc, I_C = 25 A dc				
Test 2 (Both devire types)		V_{CE} = 14 V dc, I_C = 14 A dc				
Test 3 2N6437		V_{CE} = 100 V dc, I_C = 100 mA dc				
2N6438		V_{CE} = 120 V dc, I_C = 83 mA dc				
Safe Operating area (switching)	3053	Load condition C, (unclamped inductive load), (see figure 3) T_A = 25°C, R_s = 0.1 Ω, $t_r + t_f \leq 500$ ns, Duty cycle ≤ 10%				
Test 1		t_p = 5 ms, (vary to obtain I_C), R_{BB1} = 10 Ω, V_{BB1} = 20 V dc, R_{BB2} = ∞, V_{BB2} = 0, I_C = 20 A dc, V_{CC} = 50 V dc, L = 3 μH 1/				
Test 2		t_p = 5 ms, (vary to obtain I_C), R_{BB1} = 100 Ω, V_{BB1} = 10 V dc, R_{BB2} = ∞, V_{BB2} = 0, I_C = 200 mA dc, V_{CC} = 50 V dc, L = 10 mH 2/				
Safe operating area (switching)		Clamped inductive load T_A = +25°C, duty cycle ≤ 5%, t_p = 1.5 ms (vary to obtain I_C), V_{CC} = 50 V dc, I_C = 25 A dc, (see figure 4),				
2N6437		Clamp voltage = 100 V dc				
2N6438		Clamp voltage = 120 V dc				
Electrical measurements		See table II, steps 1 and 3				

See footnote at end of table.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500

TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current 2N6437 2N6438	3041	Bias condition A, V_{BE} = 1.5 V dc, V_{CE} = 100 V dc V_{CE} = 120 V dc	I_{CEX1}		5.0	µA dc
2.	Collector to emitter cutoff current 2N6437 2N6438	3041	Bias condition A, V_{BE} = 1.5 V dc, V_{CE} = 100 V dc V_{CE} = 120 V dc	I_{CEX1}		10.0	µA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 10 A dc, Pulsed (see 4.5.1)	h_{FE2}	30	120	
4.	Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 10 A dc, Pulsed (see 4.5.1)	Δh_{FE2}	±25%		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

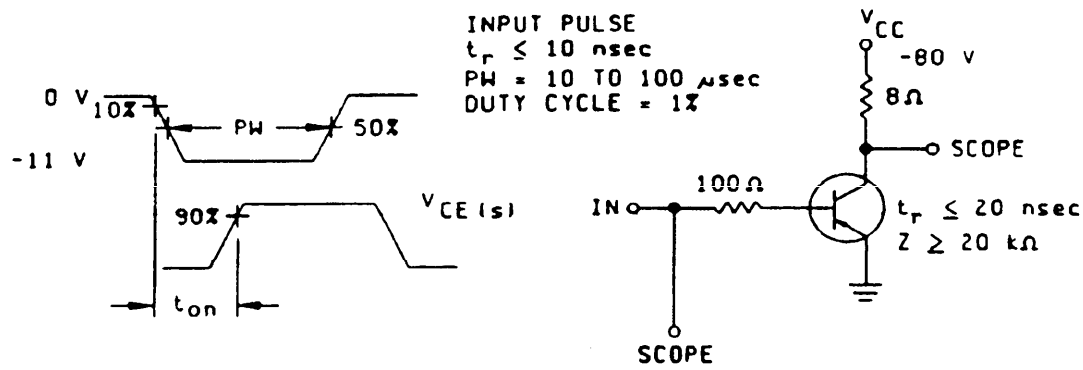
a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroups 3 and 6, see table II herein, steps 2, 3, and 4.

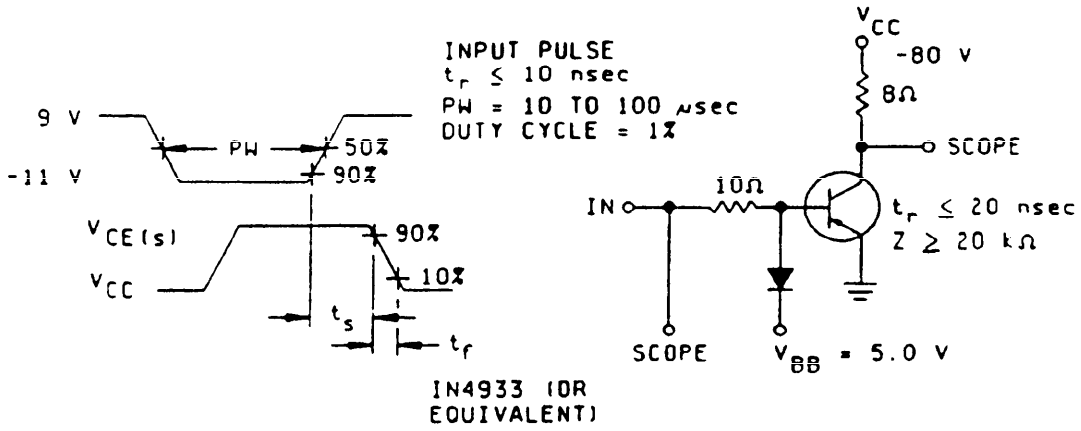
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 3.

b. Subgroup 6, see table II herein, steps 2, 3, and 4.



TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 1. Switching time te · circuits.

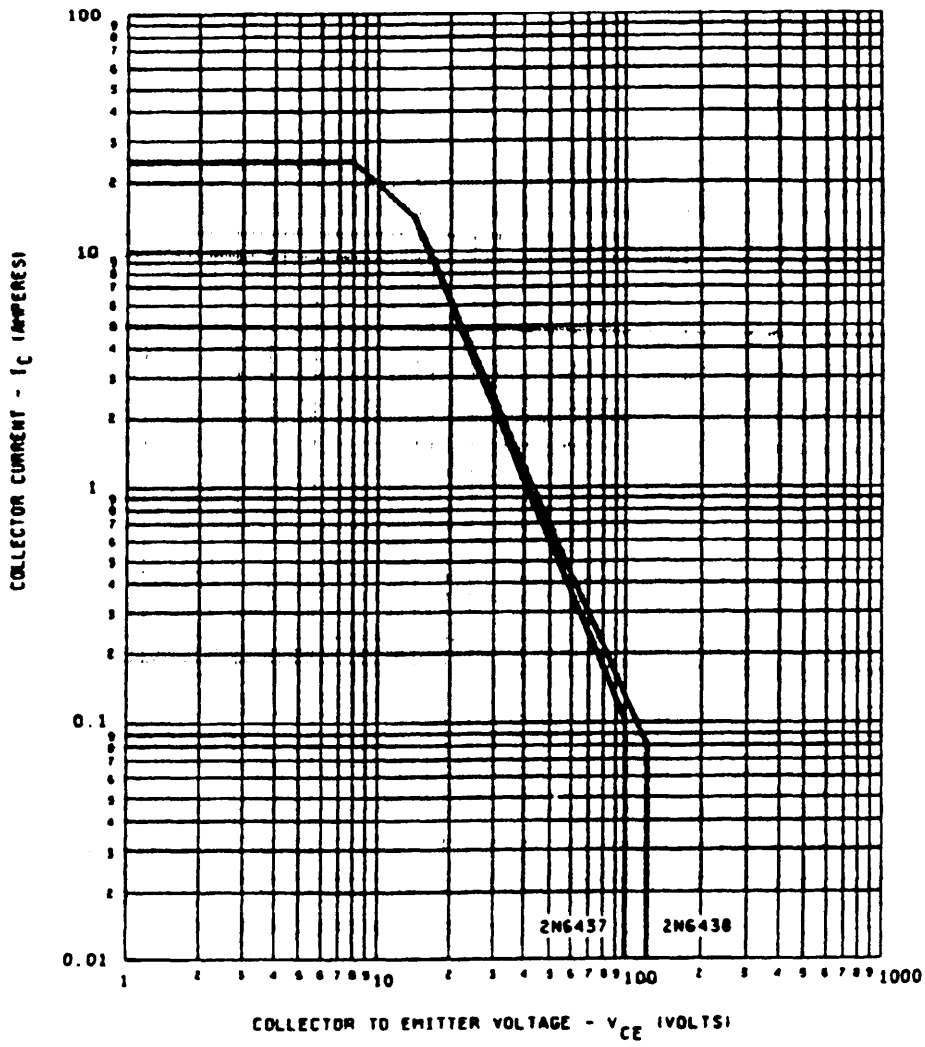


FIGURE 2. Maximum safe operating area (continuous dc).

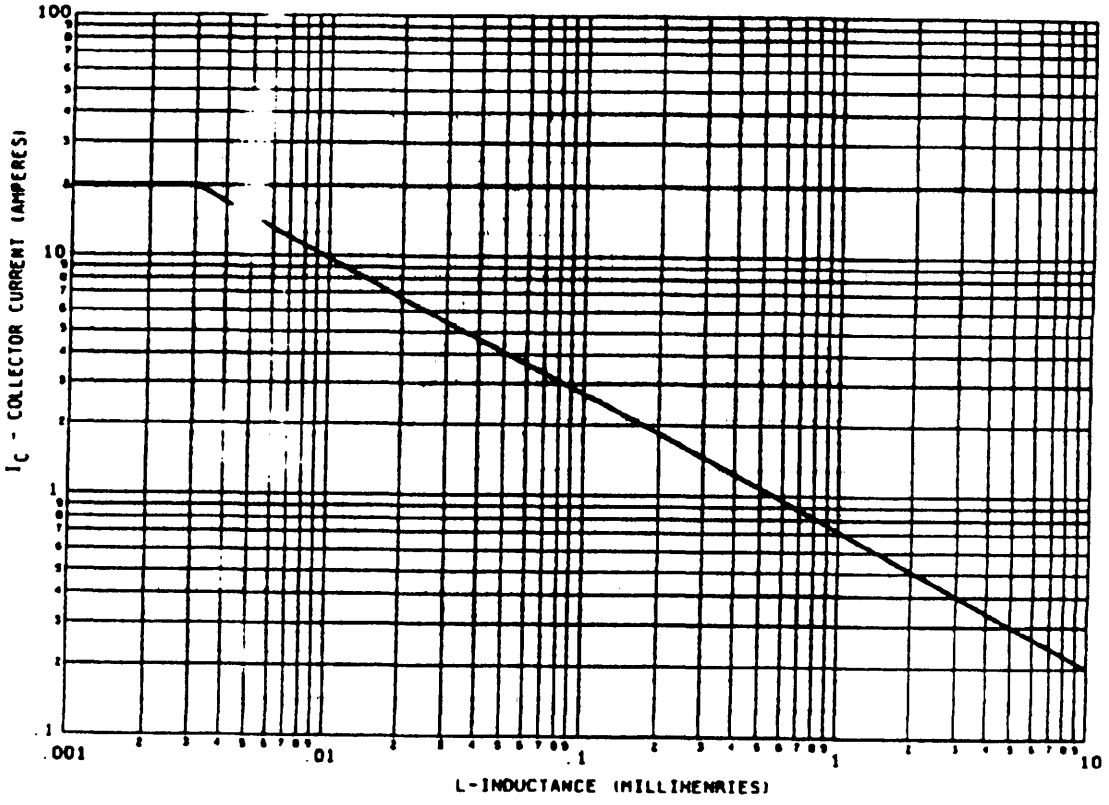
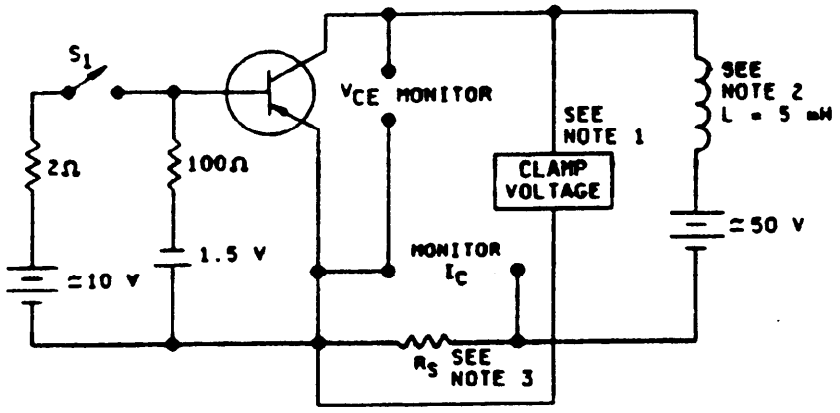


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).



NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 5 mH at 25A with a maximum dc resistance of .1 ohm. For reference only: 4 Triad C-48U; (20 mH windings in parallel) or equivalent (see 4.4.4).
3. $R_S \leq .1$ ohm, 12 W, 1 percent tolerance maximum, (noninductive).

Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage is not reached and maintained until the current returns to zero.
3. Perform specified endpoint tests.

FIGURE 4. Clamped inductive sweep test circuit.

4.5.2 Thermal resistance Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply: $R_{\theta JC}$ = .875°C/W.

a.	I_M	20 mA.
b.	V_{ce}	10 V dc.
c.	I_H collector heating current	4.0 A min.
d.	V_H collector-emitter heating voltage	10 V dc.
e.	t_h heating time	1 s (min).
f.	t_{md} measurement delay time	50 - 80 µs.
g.	t_{sw} sample window time	10 µs maximum.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specifi sation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish may be specified (see 3.3.1).

b. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Air Force - 19, 85, 99
DLA - ES
Army - AR, MI

User Activities:
Navy - AS, CG, MC, SH

Preparing activity:
Army - ER

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/508A	2. DOCUMENT DATE (YYMMDD) 931130

3. DOCUMENT TITLE
Semiconductor Device, Transistor, PNP, Silicon, Power Types 2N6437 and 2N6438, JANTX and J

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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inner conductor (seven strands of silver-plated